

Nick Kashani Motlagh

Machine Learning Engineer | Ph.D. in Computer Science and Engineering

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Available now | Open to relocating to the San Francisco Bay Area or New York City

Python/PyTorch engineer building ML training, data, and evaluation pipelines: LoRA fine-tuning, retrieval-augmented QA, and multi-GPU Slurm jobs. Four first-author peer-reviewed papers and an ISVC 2022 Springer Best Paper Award.

Technical Skills

Engineering: Python, Bash, TypeScript, PyTorch, NumPy, Hugging Face Transformers, Git, Linux, uv

ML systems: Distributed multi-GPU training, Slurm, Singularity, vLLM, LoRA/PEFT, FAISS

Methods: LLM evaluation (paired outcomes, LLM-as-judge), RAG (DPR, BM25, MonoT5), calibration, selective prediction

Experience

Computer Engineer II

May 2025 – Present

DCS Corporation / Air Force Research Laboratory

Dayton, OH

- Built **evidence-grounded QA systems**: small open models answer from retrieved documents, cite the supporting passage, and abstain when evidence falls short; LoRA on teacher-model synthetic data, paired base-vs-adapted evaluation.
- Built paired-outcome evaluation pipelines for retrieval-augmented question answering across **25,870 held-out questions** from NQ-Open, TriviaQA, and PopQA, distinguishing answers that retrieval preserves, repairs, or harms.
- Developed and evaluated LoRA fine-tuned models for answer/refine/abstain decisions using **PyTorch, Hugging Face, and Slurm**; compared learned policies with confidence-based and retrieval baselines.

Graduate Research Associate

Aug 2021 – Aug 2026

Computer Vision Lab, Ohio State University

Columbus, OH

- Implemented **PyTorch** per-class abstention thresholds (**ISVC 2022 Springer Best Paper Award**; MVA 2025 extension): CIFAR-100 selective accuracy 88.3% → 97.8% at 77.3% coverage. Released public reject-option code and calibration utilities (histogram binning, temperature scaling).
- Built contrastive metrics for multimodal translation that separate visual understanding from the translation decision: the best system reads the image correctly 81% of the time but changes its translation preference only 63% (**WMT 2024**).

Research Intern (graduate and undergraduate)

Summers 2020–2024

Air Force Research Laboratory

Dayton, OH

- Adapted and trained self-supervised vision transformers (JEPA/MAE) using distributed, containerized GPU workflows for multimodal representation learning in low-data settings (2024).
- Developed Reject Option Beam Search for machine translation (2023) and an end-to-end training procedure for naturally constrained reject-option classification (2022).
- Built an OpenStreetMap-guided Python pipeline for temporal satellite imagery collection, supporting dynamic-region analysis and an **ICCV Workshop 2021** publication.

Public Code and Demo

- 2026 re-run** of my reject-option method (code): linear probes on **4 frozen backbones** (DINOv2, DINOv3, SigLIP 2) with logits from a local RTX 4090 and a NumPy B-CDF checked by 37 parity tests. Ported to TypeScript for a live demo on nmotlagh.github.io.

Education

Ohio State University

Columbus, OH

Ph.D., Computer Science and Engineering

Conferred Aug 2026

Dissertation: *Answering Under Uncertainty: Abstention, Ambiguity, and Recoverability*

M.S., Computer Science and Engineering, May 2025 | B.S., CSE with Honors, May 2021

Selected Publications

First author on all four peer-reviewed papers. Full list: nmotlagh.github.io.

- Naturally Constrained Reject Option Classification*. Machine Vision and Applications, 2025.
- Assessing the Role of Imagery in Multimodal Machine Translation*. WMT, 2024.
- Learning When to Say “I Don’t Know”*. ISVC, 2022. **Springer Best Paper Award**.